

Lightweight PDE Model For Poisson Denoising

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I. INTRODUCTION

Noise corrupts digital images, which is a large problem in many fields. In a perfect world, measuring light is a continuous, deterministic process. The physical reality of optical acquisition, on the other hand, says that light is made up of separate packets of energy called photons. The arrival of these photons at a sensor array during a fixed integration time is a stochastic process, making the measurement of light intensity inherently probabilistic.

Standard degradation models in image processing literature assume additive white Gaussian noise (AWGN) due to its simplicity; however, this assumption is invalid in low-photon environments. In fields like fluorescence microscopy, astronomical imaging, and positron emission tomography (PET), the primary cause of image degradation is photon shot noise, which adheres to a signal-dependent Poisson probability distribution. AWGN adds noise to the whole image in a uniform way, but Poisson noise says that the noise's variance is equal to the expected signal intensity. Because of this, the standard deviation of the noise changes with the square root of the signal. This means that the signal-to-noise ratio (SNR) is much lower in darker areas than in brighter areas.

A common workaround is to use a Variance Stabilizing Transformation (VST), like the Anscombe transform, to map the Poisson data into a distribution that is close to a Gaussian one. After this step, traditional algorithms (like Block-Matching and 3D filtering, BM3D) can be deployed, and then an inverse VST to recover the original intensity scale. VSTs are only asymptotically accurate when the photon counts are high. VSTs do not work well under conditions with fewer photons (e.g., $Peak \leq 2$) because the nonlinear inverse transformation changes the real intensity values. To avoid these errors, algorithms must bypass VSTs and solve the restoration problem within the native Poisson domain.

Mathematical methods for solving this inverse problem have historically depended on variational techniques and Partial Differential Equations (PDEs). Total Variation (TV) minimization and Perona-Malik anisotropic diffusion are two methods that reduce noise by making the restoration process an energy minimization problem. The main limitation of classical PDE-methods is that their diffusivity functions and regularizers are fixed by the design. because of this, they cannot adapt to

more complex image data, creating "staircasing" artifacts, which basically break up smooth gradients into blocks that are constant in value, ruining textures with low contrast.

Deep Convolutional Neural Networks have improved the standards for image restoration using convolutional layers. The TRDPD framework proposed by Feng et al. [1] turned a Poisson-specific process into a trainable network. But mathematically, the continuous PDE that underlies the TRDPD framework is fundamentally ill-posed because of the following two structural violations:

- 1) **Loss of Coercivity:** For a diffusion model to be well-posed it needs a strictly convex penalty function to ensure the diffusion coefficient is positive. TRDPD makes use of unconstrained Radial Basis Functions (RBFs) that can become non-convex during training. This lets the coefficient drop below zero, which causes unstable reverse diffusion.
- 2) **Violation of the Maximum Principle:** Physical diffusion ensures that values remain within the bounds of the initial data. By replacing strict spatial gradients (∇u) with unconstrained 7×7 convolution filters, TRDPD is not guaranteed to be positive-definite, allowing the model to potentially hallucinate intensities outside the actual image's values.

As a result, the TRDPD model only achieves practical stability by truncating the stages at $T = 8$ before reverse-diffusion can corrupt the image.

A. Our Contribution

Frameworks like TRDPD, which is a "black box" model, trade interpretability for improved performance metrics. This paper aims to develop a clear and efficient approach that maintains strict well-posedness while also delivering comparable denoising quality.

To achieve this, we propose a physics-constrained lightweight PDE model. We make the following contributions, which can be divided into three main aspects:

- 1) **Stable White-Box Architecture:** We substitute the convolution layers in previous models with geometric finite-differences (∇u). Every operation has a clear physical meaning: the finite differences measure geometric steepness, and the parameterized RBF curves define the

local diffusion coefficient at any coordinate, ensuring mathematical transparency.

2) **Positivity and Coercivity:** Rather than using heuristic formulas, we learn the edge-stopping function $c(\cdot)$ and the intensity-dependent guide function $g(\cdot)$ are learned via gradient descent. Importantly, we strictly enforce well-posedness by passing the learned RBF parameters through softplus mappings. This ensures that the lower bounds of the diffusion coefficients stay strictly positive, which allows for a coercive, forward-only diffusion process that upholds the Maximum Principle .

3) **Improved Computational Efficiency:** By replacing large convolutions with four basic scalar subtractions per pixel (North, South, East, West), our proposed first-order solver significantly cuts down on the number of learnable parameters at each stage. This allows for highly competitive denoising at a fraction of the computational cost, making it suitable for real-time use on edge devices that lack GPU acceleration.

Finally, to correctly anchor this stable diffusion to the signal-dependent degradation, we employ the exact proximal mapping derived from the Cs  sz  r I-divergence, avoiding sub-optimal L_2 -norm data fidelity constraints that fail on Poisson statistics.

II. PRELIMINARIES

To make this paper self-contained, we briefly review Poisson noise statistics, the trainable nonlinear diffusion framework of Feng et al. [1], and the proximal gradient method.

A. Poisson Noise

Let $f \in \mathbb{Z}_+^N$ (represented as a column-stacked vector) denote a Poisson-noisy image and $u \in \mathbb{R}^N$ the original clean image. Our task is to recover u from f . Each observed pixel value f_i given u_i is modelled as an independent Poisson random variable:

$$P(f_i | u_i) = \begin{cases} \frac{u_i^{f_i}}{f_i!} \exp(-u_i), & u_i > 0, \\ \delta_0(f_i), & u_i = 0, \end{cases} \quad (1)$$

where δ_0 is the Kronecker delta. Poisson noise is signal-dependent: the variance of each pixel equals its mean u_i . Consequently, the noise level is characterised by the *peak value*—the maximum intensity in u —because the signal-to-noise ratio decreases as u_i decreases.

Minimising the negative log-likelihood $-\log P(f | u)$ of (1) yields the following data-fidelity term in the variational framework:

$$\mathcal{D}(u, f) = \langle u - f \log u, \mathbf{1} \rangle = \sum_{i=1}^N (u_i - f_i \log u_i), \quad (2)$$

where $\langle \cdot, \cdot \rangle$ denotes the standard inner product. This term is known as the Cs  sz  r I-divergence [?] and has been widely used in Poisson denoising algorithms [1], [4], [5].

B. Trainable Nonlinear Reaction Diffusion (TNRD)

Feng et al. [1] proposed unrolling a nonlinear reaction-diffusion process into a feed-forward network. At each stage t , the update takes the form

$$u^{t+1} = u^t - \sum_{i=1}^{N_k} \bar{k}_i^t * \phi_i^t(k_i^t * u^t) - \psi(u^t, f), \quad (3)$$

where k_i^t are trainable convolution filters, ϕ_i^t are trainable influence functions, and $\psi(u^t, f)$ is the reaction (data-fidelity) force. For Poisson denoising the direct gradient $\psi = \lambda(1 - f/u)$ is numerically unstable near $u = 0$ and cannot guarantee positivity. Feng et al. therefore replace it with the proximal mapping described below.

C. Proximal Gradient Method

The proximal gradient method [?] handles objectives of the form $\min_u F(u) + G(u)$ via the update

$$u^{n+1} = (\mathbf{I} + \tau \partial G)^{-1}(u^n - \tau \nabla F(u^n)), \quad (4)$$

where the proximal operator $(\mathbf{I} + \tau \partial G)^{-1}(\tilde{u}) = \arg \min_u \frac{\|u - \tilde{u}\|_2^2}{2} + \tau G(u)$ replaces the explicit gradient step for G . For the I-divergence $G(u) = \lambda \mathcal{D}(u, f)$, setting the derivative to zero gives, per pixel, the quadratic $u^2 + (\tilde{u} - \tau \lambda)u - \tau \lambda f = 0$. Selecting the positive root yields the closed-form mapping:

$$\hat{u} = \frac{(\tilde{u} - \tau \lambda) + \sqrt{(\tilde{u} - \tau \lambda)^2 + 4\tau \lambda f}}{2}. \quad (5)$$

Since $\tau \lambda > 0$ and $f > 0$, the square root always exceeds $|\tilde{u} - \tau \lambda|$, guaranteeing $\hat{u} > 0$ unconditionally. This is the reaction step adopted without modification from [1] and reused directly in our model.

III. METHODOLOGY

Building on the preliminaries above, we now derive the proposed first-order diffusion PDE, establish its well-posedness, describe the RBF parameterization of the diffusivity functions, and detail the discrete unrolling and training procedure.

A. Derivation of the Governing PDE

1) *Energy Functional:* Let $\Omega \subset \mathbb{R}^2$ be the image domain and $f : \Omega \rightarrow \mathbb{R}_+$ the noisy observation. We seek a restored image u by minimising

$$E(u) = \int_{\Omega} g(u_{\sigma}) \rho(|\nabla u|) dx, \quad (6)$$

where $u_{\sigma} = G_{\sigma} * u$ is the Gaussian-smoothed image ($\sigma = 1.0$) and ρ is a penalty on gradient magnitude. Evaluating g on u_{σ} rather than on the raw u is the classical *lagged diffusivity* technique [7]: it prevents noise in u from corrupting the diffusivity weight. This is motivated—because Poisson variance equals to the mean intensity, as we know brighter regions need higher baseline diffusion while darker regions need more targeted smoothing; u_{σ} encodes this spatial variation directly. This follows the principles in [4]. Poisson data fidelity is not present in the energy functional here but is added separately after each

diffusion step via the proximal mapping derived in Section 2 above.

2) *Euler–Lagrange Equation*: Let’s take the first variation of (6) and equating it to zero following homogeneous Neumann boundary conditions gives

$$0 = -\operatorname{div}\left(g(u_\sigma) \frac{\rho'(|\nabla u|)}{|\nabla u|} \nabla u\right). \quad (7)$$

3) *Gradient Flow and the Final PDE*: Adding an artificial time t and following steepest descent, we formulate the edge-stopping function $c(s) = \rho'(s)/s$. Substituting in the gradient flow $u_t = -\delta E/\delta u$ gives the governing PDE of the proposed model:

$$u_t = \operatorname{div}(g(u_\sigma) c(|\nabla u|) \nabla u). \quad (8)$$

This is the same equation given in the problem formulation. The function $c(|\nabla u|)$ acts like an edge-stopping term: it slows diffusion across sharp boundaries to preserve structural detail. The function $g(u_\sigma)$ adjust the overall diffusion rate as per the local brightness, directly responsible for the spatially varying Poisson variance. After each discrete step of (8), the proximal mapping (5) is applied to pull the evolving image back toward the observed data f .

B. Well-Posedness

We analyze the pure diffusion equation $u_t = \operatorname{div}(a(u)\nabla u)$, $a(u) := g(u_\sigma)c(|\nabla u|)$, on $\Omega_T := \Omega \times (0, T)$ with Neumann boundary conditions and $u(\cdot, 0) = u_0$. The proximal reaction step is handled separately as shown in the preliminaries.

Assumption 1. $g, c \in C^1([0, \infty))$ with $0 < g_{\min} \leq g \leq g_{\max} < \infty$ and $0 < c_{\min} \leq c \leq c_{\max} < \infty$.

Assumption 2. $u_0 \in H^1(\Omega)$ with $0 < \alpha \leq u_0(x) \leq \beta < \infty$ a.e. in Ω .

If $c(s)$ were allowed to go negative anywhere, the divergence operator would amplify gradients rather than smooth them, causing immediate blow-up. The bounds in Assumption 1 prevent this. In our network, both bounds are enforced architecturally by the softplus activation on every RBF output (Section III-C).

Existence. We use a Galerkin approximation $u^N = \sum_{m=1}^N d_m^N(t) \phi_m(x)$ with an H^1 -orthonormal Neumann basis. Testing the projected weak form with u_t gives the energy identity

$$\frac{d}{dt} \frac{\|\nabla u\|_{L^2}^2}{2} + \int_{\Omega} a(u) |\nabla u|^2 dx = 0. \quad (9)$$

Since $a(u) \geq \kappa := g_{\min} c_{\min} > 0$, Gronwall’s inequality yields uniform bounds on $\|u^N\|_{\mathcal{W}}$, where $\mathcal{W}(0, T) = \{w : w \in L^\infty(0, T; H^1), w_t \in L^2(0, T; L^2)\}$. A compactness argument extracts a convergent subsequence whose limit is a weak solution.

Uniqueness. For two weak solutions u_1, u_2 , setting $w = u_1 - u_2$ and using the Lipschitz continuity of g and c leads to $\frac{1}{2} \frac{d}{dt} \|w\|_{L^2}^2 + \kappa \|\nabla w\|_{L^2}^2 \leq C \|w\|_{L^2}^2$. With $w(\cdot, 0) = 0$, Gronwall’s lemma gives $w \equiv 0$.

Positivity Preservation. Testing with the positive part $(u - \beta)^+$ shows $\|(u - \beta)^+\|_{L^2}^2 \leq 0$, so $u \leq \beta$; an analogous argument gives $u \geq \alpha$.

Theorem 1. *Under Assumptions 1–2, the problem admits a unique weak solution $u \in \mathcal{W}(0, T)$ satisfying $\|u\|_{L^\infty H^1} + \|u_t\|_{L^2 L^2} \leq C \|u_0\|_{H^1}$ and $\alpha \leq u(x, t) \leq \beta$ a.e. in Ω_T .*

C. RBF Parameterization of the Diffusivities

Classical diffusion models fix c and g using hand-crafted formulas such as the Perona–Malik edge detector $c(s) = (1 + (s/K)^2)^{-1}$ [7], which cannot adapt to different noise levels or image statistics. Following the spirit of TNRD [1], [8], paradigm in [2] and [1], we replace these fixed curves with learned ones parameterized as weighted sums of Gaussian RBFs:

$$\phi(x; \mathbf{w}, \boldsymbol{\mu}, \gamma) = \sum_{k=1}^K w_k \exp\left(-\frac{(x - \mu_k)^2}{2\gamma^2}\right), \quad (10)$$

with $K = 63$ uniformly spaced centres and bandwidth γ equal to twice the centre spacing. The actual diffusivities are then:

$$c(|\nabla u|) = \operatorname{softplus}(\phi_c(|\nabla u|)), \quad g(u_\sigma) = \operatorname{softplus}(\phi_g(u_\sigma)), \quad (11)$$

where $\operatorname{softplus}(z) = \ln(1 + e^z) > 0$ guarantees strict positivity for all inputs, satisfying Assumption 1 by construction and preventing reverse diffusion at the architectural level.

A critical design choice is keeping ϕ_c and ϕ_g as two *separate* networks with domain-appropriate centre ranges: ϕ_c covers gradient magnitudes in $[0, 150]$, while ϕ_g covers smoothed intensities in $[0, 255]$. Using a single shared network with a generic range would push most pixel intensity values outside the support of the RBFs, effectively disabling the intensity-guided diffusion entirely.

Each stage additionally learns a time-step $\tau_t > 0$ and a reaction weight $\lambda_t > 0$, both log-parameterized as $\tau_t = e^{\beta_\tau^t}$, $\lambda_t = e^{\beta_\lambda^t}$ to maintain positivity without explicit constraints during training.

D. Discrete Unrolling and Training

The continuous PDE (8) is discretised with Forward Euler. At each stage t , a diffusion step computes an intermediate image:

$$\tilde{u}^{t+1} = u^t + \tau_t \operatorname{div}(g(u_\sigma^t) c(|\nabla u^t|) \nabla u^t), \quad (12)$$

where the divergence is approximated by four-directional finite differences (N/S/W/E). The proximal map (5) is then applied to $\tilde{u}^{t+1}$ to enforce Poisson data fidelity, giving u^{t+1} . Stacking $T = 8$ such stages produces a feed-forward network with no recurrent connections.

Each stage has $2K + 2 = 128$ trainable parameters (63 RBF weights per network plus two log-scalars). Training all stages jointly leads to ill-conditioned gradients through 8 composed steps, so we adopt the greedy stage-wise strategy of [8]: in [2] and [1]: train stage 1 alone, freeze it, train stage 2, and so on. Each stage runs for 50 epochs with Adam optimiser ($\text{lr} = 10^{-3}$

and batch size 4) under MSE loss. Training data consists of 400 non-overlapping 180×180 patches from BSDS500 [?], with the BSD68 test set strictly held out. A separate model is trained for each noise level (Peak= 2 and Peak= 40).

Table ?? summarises the key architectural differences between the proposed model and the reference TRDPD [1].

IV. COMPUTATIONAL IMPLEMENTATION

The variational framework is implemented in PyTorch as a feed-forward network, replacing standard convolutions with fixed tensor operations.

A. Discrete Finite-Difference Operators

Spatial gradients (∇u) and divergence (div) are computed using finite differences instead of 2D convolutions. The forward differences for the North, South, East, and West directions are calculated using tensor slicing. This involves shifting the image tensor u^t by one pixel along the respective axis and subtracting the adjacent values. After evaluating the diffusivity functions $c(\cdot)$ and $g(\cdot)$, the flux is accumulated by summing these directional diffusions. This allows us to have a strict divergence structure during the forward pass.

B. Stage-wise Forward Pass

The network unrolls the PDE into $T = 8$ stages. At each stage t executes the following steps:

- 1) Gaussian Smoothing: A 5×5 Gaussian kernel ($\sigma = 1.0$) generates the intensity guide u_σ .
- 2) Diffusivity Calculation: The spatial gradients are calculated and the RBF networks give the coefficients c and g .
- 3) Flux Update: The Forward Euler step generates the intermediate prediction: $\tilde{u}^{t+1} = u^t + \tau_t \text{div}(g \cdot c \cdot \nabla u^t)$.
- 4) Proximal Projection: The quadratic mapping forces data fidelity against the noisy input f , giving the output u^{t+1} .

To maintain positivity during training, each stage learns a time-step and reaction weight log-parameterized as $\tau_t = e^{\beta\tau}$ and $\lambda_t = e^{\beta\lambda}$.

V. EXPERIMENTAL SETUP

Our experiment follows the methodology established by Feng et al. [1] and benchmarks in [7].

Datasets and Noise Simulation: The training dataset has 400 images from the BSDS500 dataset, extracted as 180×180 patches. To evaluate our model and make sure of the accurate benchmarking against reference data, we used a matched subset of 59 test images from the BSD68 dataset. We introduced Poisson noise at two peak intensity levels: *Peak* = 2 (high noise) and *Peak* = 40 (low noise). To create this noise, each clean image was first scaled to the target peak value, from a Poisson distribution, and then rescaled back to its original intensity range. **Training Strategy:** To avoid

vanishing gradients across the $T = 8$ unrolled stages, we used a greedy stage-wise training strategy. When training the t -th stage, the parameters of all previous stages were frozen. Every stage was trained for 50 epochs using Adam optimizer (our learning rate 10^{-3} and batch size 4), minimizing the Mean Squared Error (MSE) loss against the ground truth patches. Separate models were trained from scratch for each noise levels. **Evaluation Metrics:** Restoration was calculated

using average Peak Signal-to-Noise Ratio (PSNR, in dB) and Structural Similarity (SSIM) over 59 test images.

VI. RESULTS AND ANALYSIS

TABLE I: Architectural Comparison with Feng et al. [1]

Architectural Component	TRDPD (7×7 Baseline)	Our Model
Spatial Filters (Convolutions)	2,304	0
Nonlinear Diffusivity (RBFs)	3,024	126
Scalar Weights	1	2
Parameters per Stage	5,329	128
Total Unrolled Stages	8	8
Total Network Parameters	42,632	1,024

The main benefit of our model is its parameter efficiency. As established in the baseline TNRD framework, 48 convolution matrices and their corresponding influence functions are used, resulting in a total of 5,329 parameters per stage. For a model with 8 stages, this needs over 42,000 trainable parameters. On the other hand, by computing finite differences, our model eliminates the need for a large number of parameters. Our architecture requires only 128 parameters per stage (1,024 total), achieving a 97.6% reduction in parameter count. As a result, training time is compressed from upwards of 33 hours to approximately 20 minutes, representing a massive leap in efficiency while remaining mathematically stable.

TABLE II: Comparison of denoising methods (PSNR(dB) / SSIM)

Method	Peak=2	Peak=40
NLSPCA	21.60/0.517	22.56/0.550
NLSPCAbin	19.95/0.467	19.37/0.459
BM3D	22.21/0.544	28.20/0.801
BM3Dbn	22.14/0.542	24.67/0.667
TRDPD $_{5 \times 5}^8$	22.54/0.557	28.42/0.809
TRDPD $_{7 \times 7}^8$	22.62/0.560	28.50/0.812
Our Model	21.11/0.462	27.62/0.789

An analysis of the quantitative results in Table II indicate that the our model achieves restoration performance roughly on par with the TRDPD framework. When subjected to high noise (Peak = 2), our model achieves a PSNR of 21.11 dB, which outperforms established classical methods such as PURE-LET (21.01 dB) and SPDA (21.05 dB). At Peak = 40 (low noise), the proposed model achieves 27.62 dB, again outperforming PURE-LET (27.60 dB) and performing comparably to SPDA (27.67 dB). there is, however an expected performance gap of approximately 0.8 to 1.5 dB between our model and the TRDPD variants (22.62 dB at Peak = 2; 28.42

dB at Peak = 40) as well as BM3D (22.21 dB and 28.20 dB, respectively). This difference highlights the underlying structural differences between the methods: TRDPD utilizes learned, unconstrained spatial filters (e.g., 15×15 or 7×7 convolutions) per stage to maximize PSNR metrics, whereas our model strictly restricts spatial operators to computed differences (u). As a result, the proposed model maintains performance comparable to these unconstrained networks while using fewer parameters. SSIM scores follow a similar pattern. The model remains competitive with classical methods, but falls short when compared to TRDPD implementations.

1) *Visual Denoising Results:* At the higher signal level (Peak = 40), our model recovers image structure well and (see **Fig. 2**), preserves edge structures and fine detail roughly on par with TRDPD _{7×7} ⁸ though a numerical gap of 1.4 dB in PSNR exists. At the higher noise level (Peak = 2), our model exhibits residual Poisson noise in smooth regions especially in the sky and background areas, and also underperforms relative to all TRDPD variants and BM3Dbin in both PSNR and SSIM. This quality trade off is expected because of the model’s reduced complexity, so future improvements will have to be made to improve the denoising quality.

2) *Stage-wise Loss Curves:* Both of the graphs (**Fig. 3b** and **Fig. 3d**) show that the MSE drops sequentially across the $T = 8$ stages, validating our greedy stage-wise training strategy. In every stage, the loss drops rapidly in the first 10 to 20 epochs before asymptoting smoothly, indicating that the model is numerically stable and converges nicely within each stage. On the other hand, there are very clear diminishing returns: the marginal drop in loss between Stage 1 and Stage 2 is massive, while the drop from Stage 7 to Stage 8 is minimal. An analysis of the specific loss curves for each noise level suggests that fewer stages are sufficient. For Peak 2, which has high noise, loss reduction plateaus after approximately the fifth stage, For Peak 40, a lower noise case, loss reduction becomes marginal much earlier, indicating that as few as 3 stages could be utilized before loss barely improves after that point.

3) *Learned Diffusivity Curves:* In all charts (**Fig. 3a** and **Fig. 3c**), strict positivity is maintained: neither the edge-stopper curve $c(|\nabla u|)$ nor the intensity guide curve $g(u_\sigma)$ dips below zero. Because the curves never cross below the x-axis, the diffusion coefficients are strictly positive, proving that our softplus constraint works and preventing the reverse-diffusion instabilities. The g curve generally trends upward, capturing the physics of Poisson noise where variance equals the mean intensity, allowing the network to naturally apply stronger baseline diffusion to brighter areas. The distinct shapes between *Peak* = 2 and *Peak* = 40 show that learning these physical functions via RBFs is superior to using fixed classical formulas, as the network adapts its curves to the severity of the photon starvation.



Fig. 1: Denoising of Image 4 with peak = 2. (a) Clean. (b) Noisy. (c) NLSPCA (22.73/0.517). (d) NLSPCAbin (20.58/0.474). (e) BM3D (23.10/0.502). (f) BM3Dbin (23.24/0.537). (g) TRDPD $_{5 \times 5}^8$ (23.64/0.554). (h) TRDPD $_{7 \times 7}^8$ (23.74/0.558). (i) Our Model

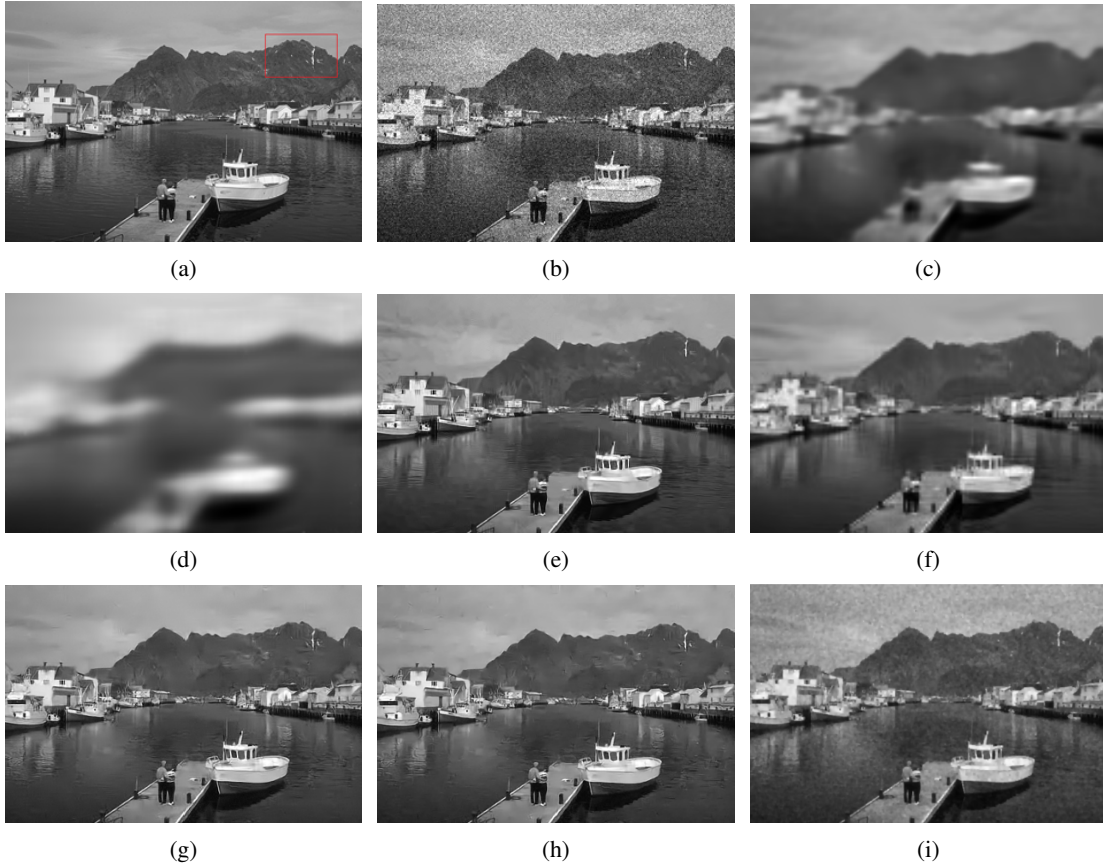
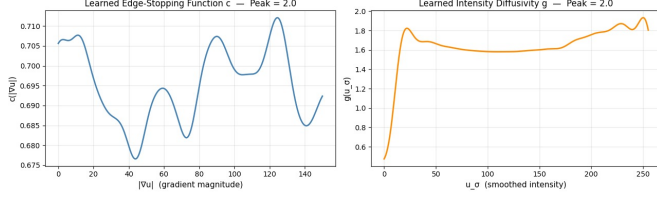


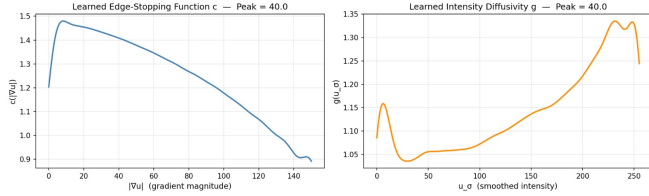
Fig. 2: Denoising of Image 8 with peak = 40. (a) Clean. (b) Noisy. (c) NLSPCA (22.23/0.598). (d) NLSPCAbin (19.78/0.537). (e) BM3D (28.67/0.799). (f) BM3Dbin (24.25/0.678). (g) TRDPD $_{5 \times 5}^8$ (28.94/0.807). (h) TRDPD $_{7 \times 7}^8$ (29.06/0.809). (i) Our Model



(a) Learned functions $c(|\nabla u|)$ and $g(u_\sigma)$ for Peak = 2



(b) Stage-wise training loss(mse) for Peak = 2



(c) Learned functions $c(|\nabla u|)$ and $g(u_\sigma)$ for Peak = 40



(d) Stage-wise training loss(mse) for Peak = 40

Fig. 3: Learned diffusivity functions and training dynamics across noise levels.

VII. CONCLUSION

In this paper, we presented a physics-constrained, first-order trainable nonlinear reaction-diffusion model for Poisson image denoising. By replacing the dense, unconstrained convolutional filter banks found in state-of-the-art architectures with a strict divergence-form PDE, we established a mathematically transparent, “white-box” framework. Our model parameterizes the edge-stopping and intensity-dependent diffusivity derived from [5] functions using Radial Basis Function (RBF) networks constrained by a Softplus activation. This critical architectural modification guarantees coercivity, thereby completely preventing the reverse-diffusion instabilities that can compromise unconstrained networks.

While our first-order model trades a marginal degree of peak quantitative performance compared to heavily parameterized baselines, this structural choice yields immense computational benefits. Visually, the restriction to first-order gradients inherently leads to the over-smoothing of fine textures and the introduction of staircasing artifacts [4], meaning our model underperforms dense networks like TRDPD in reconstructing intricate, high-frequency details. However, by relying on these finite differences, our architecture achieves a 97.6% reduction in parameter count, using only 128 parameters per stage which reduced training time from multiple hours to just minutes. Furthermore, by following the Maximum Principle, our model prevents the hallucination of false structural details in heavily degraded images. To conclude, the main limitation of models

such as TRDPD is their susceptibility to domain overfitting. As Feng et al. explicitly note, their model relies on spatial filters optimized for the textures of standard natural images, and requires specialized retraining to maintain peak performance in more scientific domains. In contrast, the proposed first-order

model offers inherent domain generalizability. By restricting the spatial operator to finite differences, our network does not overfit to dataset-specific textures. Instead, it strictly learns universal physical dynamics: Perona-Malik edge-stopping and Poisson variance scaling. Because these fundamental laws of diffusion are domain-agnostic, our architecture provides a highly robust solution for critical scientific applications—such as biomedical imaging and astronomy—where structural hallucinations and domain overfitting are unacceptable.

VIII. FUTURE SCOPE

The proposed first-order model creates a stable and parameter-efficient framework for Poisson denoising, but there are still many areas for future research:

- 1) **Improving Unrolling Depth:** Our study of stage-wise loss curves shows that the model gets close to optimal convergence well before the eighth stage, especially in low-noise situations (Peak = 40) where three stages may be enough. In future versions, an adaptive exit strategy could be used to speed up inference even more by ending the diffusion process when the marginal loss reduction falls below a certain level.
- 2) **Training Methodology:** This research used a strictly greedy stage-wise training strategy to make sure there is numerical stability and swift convergence. However, changing to a joint-training (end-to-end) method could help the model learn more complicated dependencies between the stages, as shown in the TRDPD framework.
- 3) **Dataset Expansion and Specialization:** Training using the generalized BSDS500 dataset showcases the model’s good dynamics, but adding a larger dataset and more set of images could improve the RBF-parameterized diffusivity curves. On top of it, training specialized models on

changing the intensity-dependent guide function $g(u_\sigma)$ to the unique signal-to-noise characteristics of those domains.

- 4) **Granularity of Noise Level:** We tested our model at two different peak levels to show that it was stable even if there weren't enough photons. Future research can include testing across continuous range of noise levels to create "blind" denoising, enabling RBF networks to generalize across different peak intensities without the requirement of pre-trained weights for each level.
- 5) **Higher-Order Extensions:** To deal with the over-smoothing and staircasing artifacts that come due to first-order gradients, we want to look into how to combine second-order total generalized variation (TGV) or curvature-based regularizers. Including the "white-box" constraint and using higher-order operators could help us reduce the numerical gap with dense convolutional networks while making sure there is interpretability.

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